Junyan (Ryan) Zhou

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Research Dr, Durham NC
https://junyanzhou.github.io/

BACKGROUND

Duke University Aug '25 - Present
Master of Science in Mechanical Engineering(Robotics) GPA: 4.0/4.0

General Robotics Lab at Duke University May '26 - Present
Research Assistant

Wuhan University of Technology Sep '21 - Jun '25
Bachelor of Engineering in Automotive Engineering GPA: 3.99/5.0(5%)

RESEARCH EXPERIENCES

Geometry-Robust Tri-Axis Tactile Sensing with Improved Signal Decoupling ***May 2026 - Present***
Independent Researcher, advised by Prof. Boyuan Chen

- Developed a tactile sensing system end-to-end, encompassing mechanical design, optimization, and fabrication, custom PCB design, embedded signal processing, and user interface development.
- Conducted robot-assisted sensor characterization and benchmarking against an existing tactile sensor using the Franka Research 3.
- Evaluating tactile sensor performance in robotic manipulation tasks using asymmetric actor–critic PPO and privileged-to-sensorimotor latent distillation.

External-Support-Aware Humanoid Control for Whole-Body Manipulation ***Aug 2026 - Present***
Researcher

- Developed a contact-constrained reference-generation pipeline for Unitree G1 using Crocoddyl and Pinocchio, incorporating joint limits, collision screening, and contact-force feasibility checks.
- Adapted an existing teacher–student reinforcement learning framework for Cartesian foot-reaching commands, using proprioceptive history and learned state and joint-target estimators.

OTHER EXPERIENCES

Desktop-Scale Bipedal Robot (Course Project for ME555 Robot Studio) ***Jan 2026 - Apr 2026***
Individual Contributor

- Designed and developed a desktop-scale bipedal robot end-to-end, achieving the highest walking speed and longest distance in the course's final competition.

VOYAH Automotive Technology Co., Ltd. (Intern) ***Jan 2025 - Apr 2025***
Test Engineer

- Applied digital filtering and spectral analysis to in-cabin audio signals to evaluate noise interference and speech recognition robustness.

AWARDS & SCHOLARSHIP

- Outstanding Graduate Student, '25.
- Merit Student of the University ('21 - '24).
- University 1st-class / 3rd-class Scholarship: ('21 - '25).
- Excellent Student in Sunshine Sport Program '24.

TECHNICAL SKILLS

Mechanical Engineering: CAD(SolidWorks), CAE(ABAQUS), 3D Printing (Bambu Lab), Soldering
Electrical Engineering: PCB Design, Electrical System Integration, Embedded Systems (Sensor Acquisition, Motor Control, Serial Communication)
Computer Science: Python, C++, MATLAB, ROS2, MuJoCo, Linux

OTHERS

- **Languages**: English (C1), Mandarin (Native).
